

Figure 1: A Venn diagram definition of data science by Drew Conway
(Image credit: googleimages.com)

data. This ultimately helps the system to make decisions. Let us have a look at the different processes followed in data science.

Collection of raw data: This is the first step implemented in data science and deals with the collection of actual raw data, on which different data science operations need to be performed. There are broadly two ways to do this:

1. We can pick one or many tools to collect data automatically from different data sources. This option is widely used in order to collect data from large data sources. We just need to copy-paste a small code snippet into our website and we are ready to go (e.g., Hotjar, Google Analytics, etc).
2. We can also collect the data for ourselves using a JavaScript code snippet that sends the data in a .csv plain text file on the server. This is a bit difficult to implement as it requires some coding skills. But if we think about the long term, this solution is more profitable.

Data processing: This refers to the refinement of raw data that has been collected during the data collection process. We all know that the raw data is unprocessed and unorganised. It needs to be arranged and organised so that it becomes easier to perform operations on it. Once the data is processed, we get an output data which is the processed, categorised and summarised version. Data processing is required in most of the experiments and surveys. The collected raw data sometimes contains too much data to analyse it sensibly. This is especially the case when we do research using computers as this may produce large sets of data. The data then needs to be organised or manipulated using the deconstruction technique.

Data set cleaning: This is the process of removing unwanted data from the processed data set and keeping only what's required for analysis. This helps to reduce the large set of data to a smaller one by removing the inconsistent or incorrect data, and makes it easier to perform different analysis tasks on it.

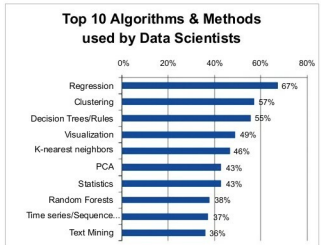


Figure 2: The top algorithms and methods used by data scientists
(Image credit: googleimages.com)

Exploratory data analysis: This approach is used to analyse the data sets in order to summarise their important characteristics, often with the help of visual methods. A statistical model can also be used for analysis, but primarily, exploratory data analysis is for visualising what the data can tell us beyond the formal modelling or the hypothesis testing task. This approach was promoted by John Tukey to encourage different data scientists to explore the data, and hence possibly formulate the hypotheses that could lead to new methods of data collection and experiments. This is different from the initial data analysis, which focuses mostly on checking the assumptions required for model fitting, the hypothesis testing, the handling of different missing values and making transformations of variables as required.

Models and algorithms: Once the data is cleansed, some sets of data will need exploratory analysis whereas other sets can be directly used for the selection of data models and algorithms. This phase of data science deals with the process of selecting the right and appropriate algorithm on the basis of the data set obtained after data cleaning, and also on the basis of the knowledge obtained about the data set during exploratory data analysis. The algorithm chosen is such that it's most efficient for the available data set. This process also includes the design, development and selection of the data models, which can be used to perform the required operations on the data, to obtain the required data product.

Report communication: This is the part of data science that deals with generating and developing visual reports in the form of graphs and pie-charts, which can be used by data scientists to analyse the data patterns and make the appropriate decisions. This decision is the final output, which is then utilised in different applications.

Data product: This is the final data product, which is used to continuously improve and change the application system whose data is analysed. This can be considered as the end product, which represents the whole set of operations performed on the collected raw data set.